Manchester Metropolitan University

MSc Dissertation Terms of Reference;

Robot Arm Manipulation using Visuomotor Diffusion Policy

1. Background and Introduction

Robot manipulation has emerged as a critical domain in robotics research, with recent advances in deep learning transforming how robots learn complex motor skills. Traditional approaches to robot control often struggle with the multimodal nature of real-world manipulation tasks, where multiple valid action sequences may exist for achieving the same goal (Chi et al., 2023). The field has witnessed significant progress with the introduction of generative models for policy learning, addressing fundamental challenges in visuomotor control (Song et al., 2025).

Diffusion Policy represents a breakthrough in robot learning, introduced by Chi et al. in 2023. This approach models robot visuomotor policies as conditional denoising diffusion processes, demonstrating remarkable improvements over existing state-of-the-art methods with an average performance gain of 46.9% across 12 different robot manipulation tasks (Chi et al., 2023). The method excels at handling multimodal action distributions, high-dimensional action spaces, and exhibits impressive training stability (Chi et al., 2023).

The core innovation of Diffusion Policy lies in its formulation of action generation as an iterative denoising process. Instead of directly outputting actions, the policy learns the gradient of the action-distribution score function and optimises through stochastic Langevin dynamics steps (Chi et al., 2023). This approach inherits key advantages from diffusion models: the ability to represent complex multimodal distributions while maintaining stable training characteristics (Ho, Jain and Abbeel, 2020).

Key technical contributions include the integration of receding horizon control, which combines action sequence prediction with closed-loop execution, visual conditioning that treats observations as conditioning rather than joint data distribution, and the time-series diffusion transformer architecture that minimises over-smoothing effects (Song et al., 2025).

The method has been successfully demonstrated on various robot platforms, including UR5 and Franka robots, performing challenging tasks such as precise pushing, object manipulation, and bimanual coordination (Chi et al., 2023).

2. Aim and Objectives

2.1 Aim

To successfully reproduce the visuomotor diffusion policy framework for robot arm manipulation on the Sawyer robot platform, with a specific focus on achieving successful completion of the custom push-T task or similar.

2.2 Objectives

1. Reproduce the core diffusion policy architecture as described in the original paper (Chi et al., 2023)
2. Adapt the implementation to work with the Sawyer robot's 7-DOF arm configuration on the push-T task or in simulation.
3. Collect demonstration data for training the diffusion policy
4. Integrate the policy with Sawyer's control system through ROS interfaces
5. Establish success criteria and evaluation metrics for task completion
6. Train the diffusion policy model using collected demonstration data
7. Evaluate policy performance on the push-T task
8. Compare results against baseline methods where applicable
9. Analyse the model's ability to handle multimodal action distributions
10. Analyse the effectiveness of different network architectures (CNN vs. Transformer)

3. Software and Hardware Requirements

3.1 Hardware Requirements

Sawyer Robot Platform:

1. Rethink Robotics Sawyer 7-axis collaborative robot arm
2. Maximum reach: 1260mm, payload capacity: 4kg
3. Embedded and mounted vision camera system

Computing Requirements:

1. **CPU**: Intel Core i7 (minimum 12 cores/24 threads)
2. **GPU**: NVIDIA RTX 4070 with minimum 8GB VRAM for training
3. **RAM**: 32GB minimum
4. **Storage**: 500 NVMe SSD for fast data loading and model storage

3.2 Software Requirements

Operating System:

1. Ubuntu 20.0 LTS (recommended for ROS1 compatibility)

Core Frameworks:

1. **ROS Noetic**: For robot control and communication
2. **PyTorch**: Version 1.8+ with CUDA support
3. **Python**: Version 3.10+ with scientific computing libraries

Deep Learning Libraries:

1. PyTorch with torchvision for neural network modelling.
2. NumPy, SciPy for numerical computation
3. OpenCV for computer vision tasks
4. Matplotlib for visualisation and analysis.

Development Tools:

1. VSCode IDE for the project development
2. Git/Github for version control
3. Jupyter notebooks for experimentation and analysis
4. Weights & Biases or TensorBoard for experiment tracking.

4. Policy Network Architecture Requirements

The proposed robot policy implementation will support both CNN-based and Transformer-based architectures as outlined in the original work.

CNN Architecture:

1. FiLM (Feature-wise Linear Modulation) conditioning for visual observations (Perez et al., 2017).
2. U-Net backbone for the denoising network
3. Suitable for tasks requiring spatial feature extraction

Transformer Architecture:

1. Multi-head attention mechanism for sequence modelling (Vaswani et al., 2017).
2. Cross-attention layers for observation conditioning
3. Preferred for tasks requiring temporal reasoning and high-frequency control

The choice between architectures will be determined based on task complexity and performance requirements, with CNN-based implementation recommended as the initial approach (Chi et al., 2023).

5. Success Criteria

The dissertation will be considered successful upon achieving:

1. **Technical Implementation**: Successful reproduction of the diffusion policy framework adapted for the Sawyer robot.
2. **Task Completion**: Demonstration of successful move T task execution with measurable performance metrics.
3. **Documentation**: Comprehensive analysis of implementation challenges, solutions, and performance evaluation.

4. **Academic Contribution**: Clear documentation of the adaptation process and lessons learned for future implementations.

This research will contribute to the growing body of work on diffusion-based policies for robot manipulation while providing practical insights into implementing state-of-the-art learning algorithms on real robot hardware.

6. Project Timeline

The table below is a suggested **3-month timeline** from **June 1 to August 29, 2025**, broken down into key project phases with deliverables.

Week(s)	Dates	Activities
Week 1	June 1 – June 7	1. Write an ethics application and terms of reference. 2. Project planning
Week 2–3	June 8 – June 21	1. Conduct a literature review 2. Study the original diffusion policy paper & code
Week 4	June 22 – June 28	1. Set up the software environment and the Sawyer hardware 2. Verify dependencies
Week 5–6	June 29 – July 12	1. Reproduce diffusion policy baseline (simulated tasks with CNN backbone for state-based model)
Week 7	July 13 – July 19	1. Adapt and integrate the model for the Sawyer robot arm 2. Integrate visual input from camera sensors
Week 8–9	July 20 – August 2	1. Implement and train on the custom push-T task 2. Evaluate performance for state-based and visual experiments.
Week 10	August 3 – August 9	1. Analyse results and model behaviour 2. Compare with baseline methods
Week 11	August 10 – August 16	1. Begin writing dissertation (Methodology, Experiments)

Week 12	August 17 – August 23	1. Finish the literature review & discussion sections
Week 13	August 24 – August 29	1. Finalise documentation & proofreading 2. Submit dissertation draft

Key Milestones

1. Ethics approval and terms of reference complete
2. Literature review completed
3. Baseline diffusion policy reproduced
4. Custom push-T task completed and evaluated
5. Final dissertation submitted

7. Reference list

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Perez, E., Strub, F., de Vries, H., Dumoulin, V. and Courville, A. (2017). FiLM: Visual Reasoning with a General Conditioning Layer. *arXiv:1709.07871 [cs, stat]*. [online] Available at: https://arxiv.org/abs/1709.07871.

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